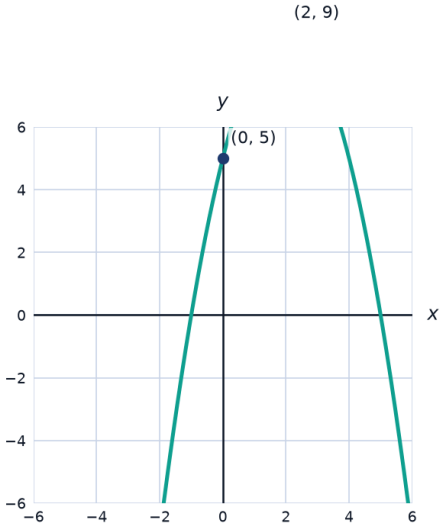
Functions & Their Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

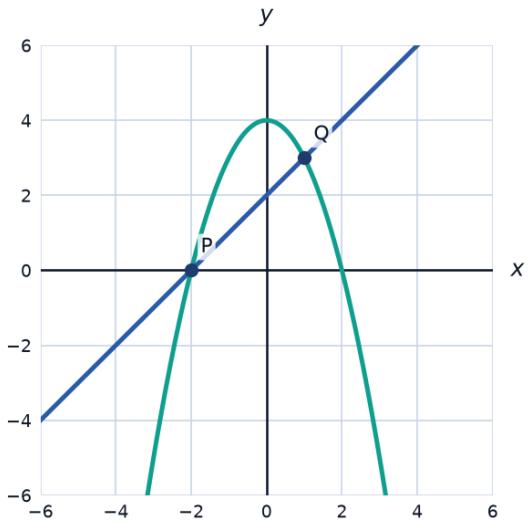
1 [Foundational] • Target: Review: Relations and functions

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



2 [Foundational] • Target: Function families and transformations

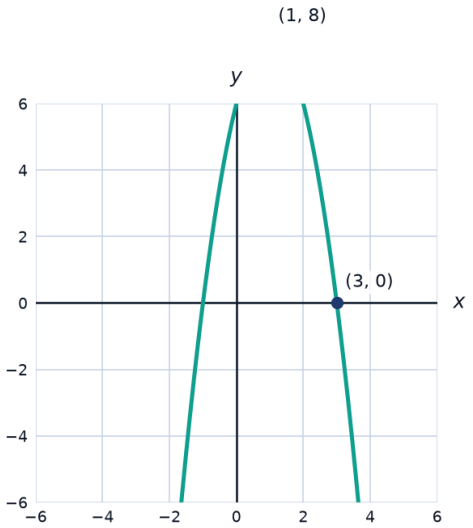
The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.



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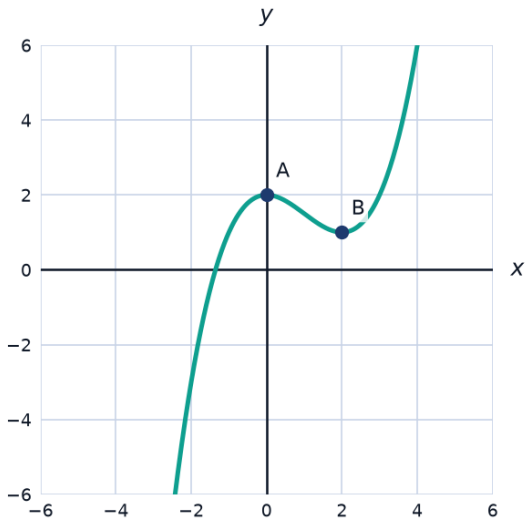
3 [Foundational] • Target: Choosing appropriate models

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



4 [Foundational] • Target: Domain and range and continuity

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?



5 [Foundational] • Target: Characteristics from a graph

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

6 [Foundational] • Target: Average rates of change

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

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7 [Foundational] • Target: Review: Relations and functions

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Function families and transformations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

9 [Intermediate] • Target: Choosing appropriate models

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

10 [Intermediate] • Target: Domain and range and continuity

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

11 [Intermediate] • Target: Characteristics from a graph

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

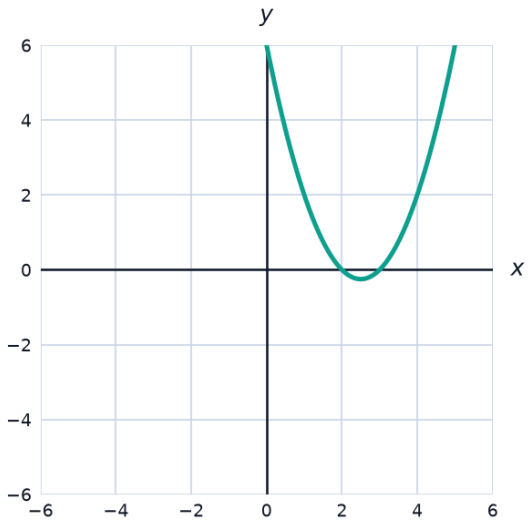
12 [Intermediate] • Target: Average rates of change

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of x is excluded from the domain?

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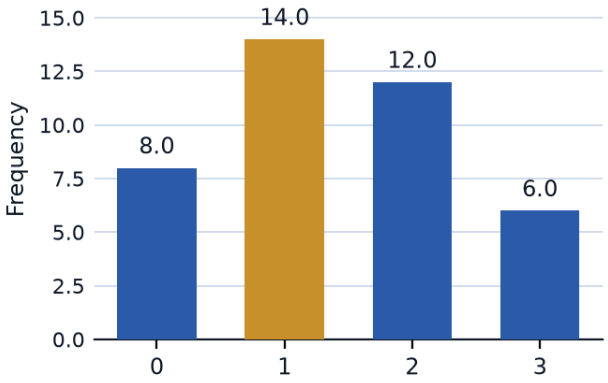
13 [Intermediate] • Target: Review: Relations and functions

Consider the rational function $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. Let $g(x)$ be the function obtained by simplifying $f(x)$. Determine the value of $g(x)$ as $x \to \infty$.



14 [Intermediate] • Target: Function families and transformations

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays k post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of k?

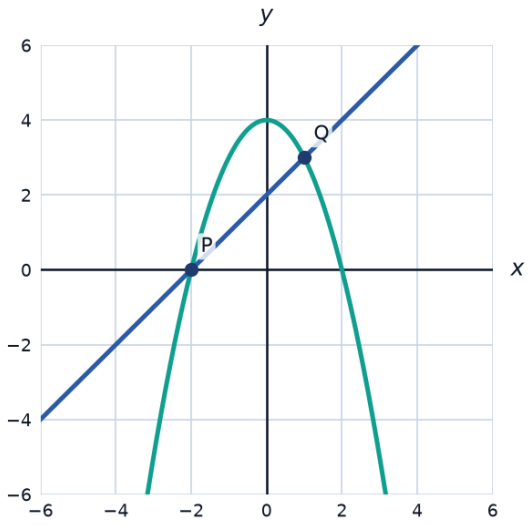


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 4.01 Rational Expressions

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?



16 [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the period of the trigonometric function $f(x) = 5\cos(4x) - 2$?

17 [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 6.01 Radicals & Rational Exponents

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

18 [Challenging] • Target: Average rates of change • Cross-Lesson: 7.01 Sequences & Series

Determine all continuous even functions f satisfying $f(f(x)) = x$.

19 [Challenging] • Target: Review: Relations and functions • Cross-Lesson: 8.01 Conic Sections

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

20 [Challenging] • Target: Function families and transformations • Cross-Lesson: 9.01 Trigonometric Functions

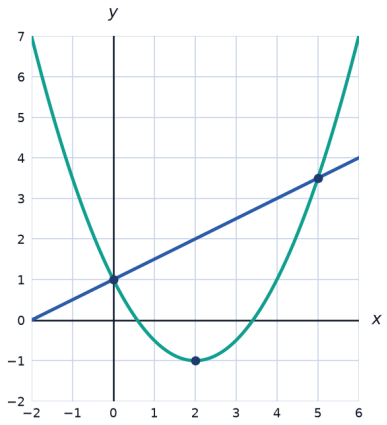
Consider the polynomial function $P(x) = 2x^3 - 5x^2 + 3x - 7$. What is the value of $P(2)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Review: Relations and functions

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(2)$.



a. 7

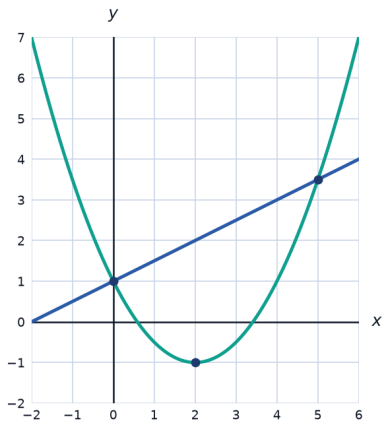
b. 9

c. 4

d. 11

22 [Foundational] • Target: Review: Relations and functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

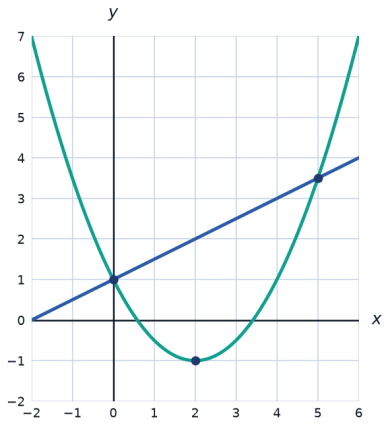
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Review: Relations and functions

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(5)$.



a. 13

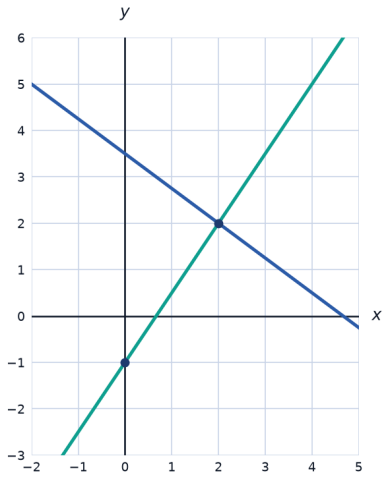
b. 15

c. 10

d. 17

24 [Foundational] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(2)$.



a. 10

b. 13

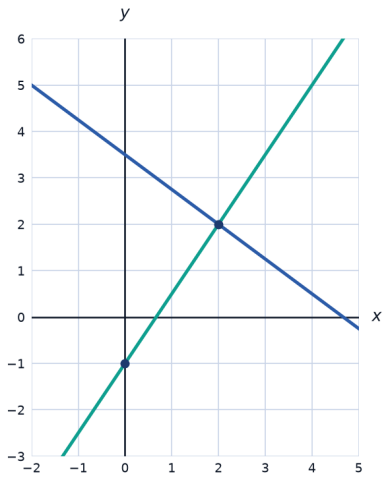
c. 6

d. 14

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25 [Foundational] • Target: Function families and transformations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

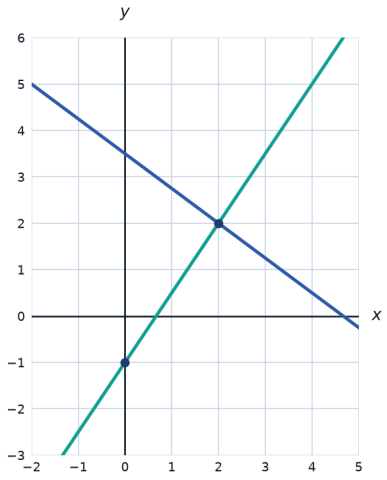
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(5)$.



a. 19

b. 22

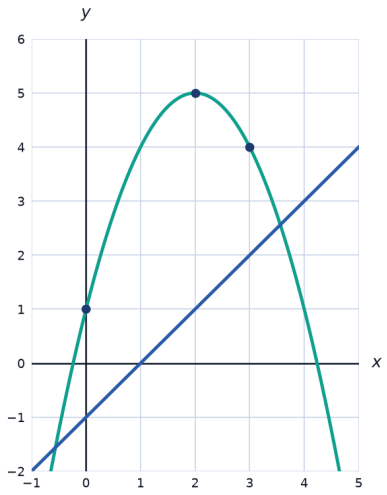
c. 15

d. 23

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27 [Foundational] • Target: Choosing appropriate models

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(2)$.



a. 13

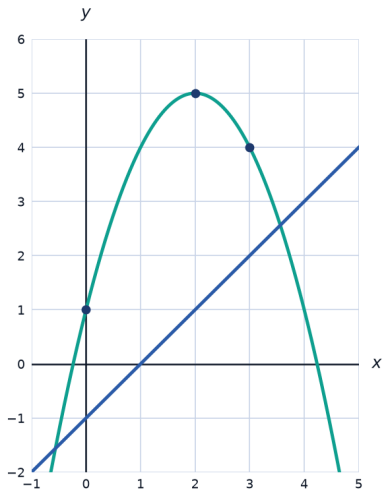
b. 17

c. 8

d. 17

28 [Foundational] • Target: Choosing appropriate models

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Review: Relations and functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Review: Relations and functions

A data set recorded for review: relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

31 [Intermediate] • Target: Review: Relations and functions

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

32 [Intermediate] • Target: Review: Relations and functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Function families and transformations

A data set recorded for function families and transformations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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34 [Intermediate] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

35 [Intermediate] • Target: Function families and transformations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

36 [Intermediate] • Target: Choosing appropriate models

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

37 [Intermediate] • Target: Choosing appropriate models

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

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39 [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for domain and range and continuity consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

41 [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 5x + 6$ in domain and range and continuity, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

42 [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 11.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

43 [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A data set recorded for characteristics from a graph consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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44 [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 6x + 7$ in characteristics from a graph, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

45 [Challenging] • Target: Average rates of change • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 2x + 8$ in average rates of change, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

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